

	1	2	3	4	5	6
A	RP235xB File: PGA2350B.kicad_sch Stepper Output	Ethernet File: Ethernet.kicad_sch Misc IO				
	File: StepperOutput.kicad_sch Spindle	File: MiscIO.kicad_sch Limit and Control Inputs				
B	File: Spindle.kicad_sch DigitalIO and Relays	File: LimitandControlInputs.kicad_sch Power				
	File: DigitalIO and Relays.kicad_sch Servo_error	File: power.kicad_sch				
	File: servo_error.kicad_sch					
C						
D						
	1	2	3	4	5	6

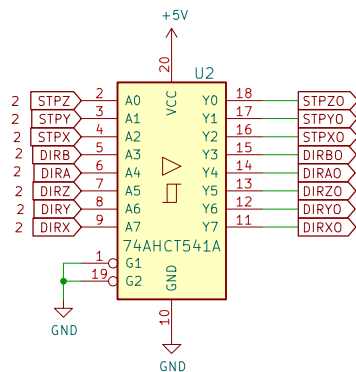
Brookwood Design

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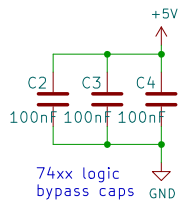
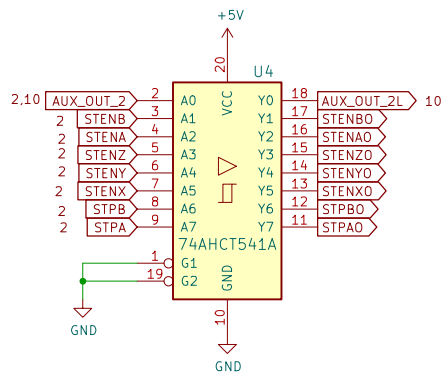
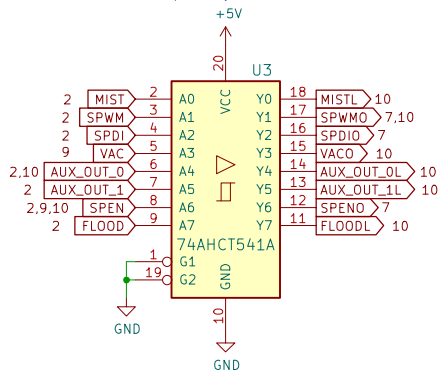
Title: RP23U5XB

Size: A4 Date: 2026-05-17
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Digital output section. 3.3V inputs, 5V outputs. Use of AHCT logic allows this. Drive capability of 8mA.



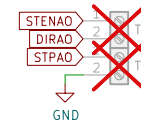
X Axis driver interface.



Y Axis driver interface.



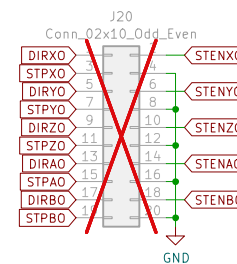
Z Axis driver interface.



A Axis driver interface.



B Axis driver interface.



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Sheet: /Stepper Output/
File: StepperOutput.kicad_sch

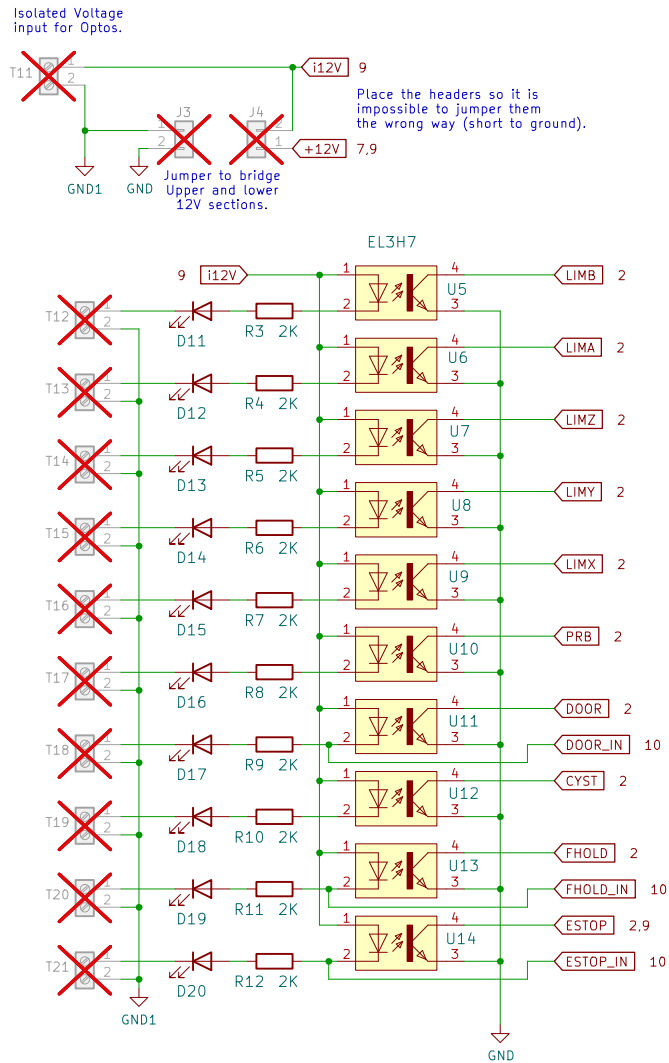
Title: RP23U5XBB

Size: A4 Date: 2025-03-27

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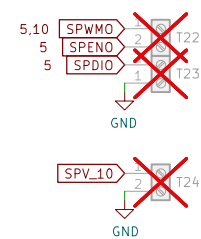
Brookwood Design

Sheet: /Limit and Control inputs/
File: LimitandControlInputs.kicad_sch

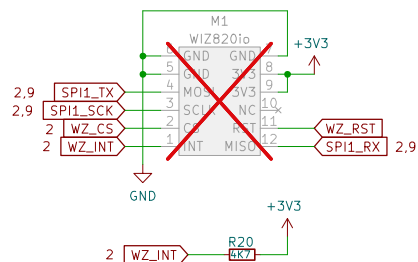
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Size: A4 Date: 2025-03-27
KiCad E.D.A. 8.0.9

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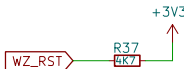
Id: 7/10



The WZ850io datasheet shows a pullup resistor on it but there have been reports of instability when WZ_INT does not have the additional pullup resistor. So, included out of caution – the additional pullup does not seem to hurt.

Wiz820io used here is technically the Wiz850io. Marketed as being compatible with the Wiz820io. It uses the W5500 chip. The Wiz820io uses the W5200 chip and is not supported. The ones called Wiz850-lite (Chinese knockoffs of the Korean original) are also supported.

Yes, it is confusing – just make sure the module has a W5500 on it and 2 6x1 pin headers on either side.



RST is held high. Need to make sure that there will not be a case where RST (active low) is needed. Evaluate if a higher value (10K) is better.

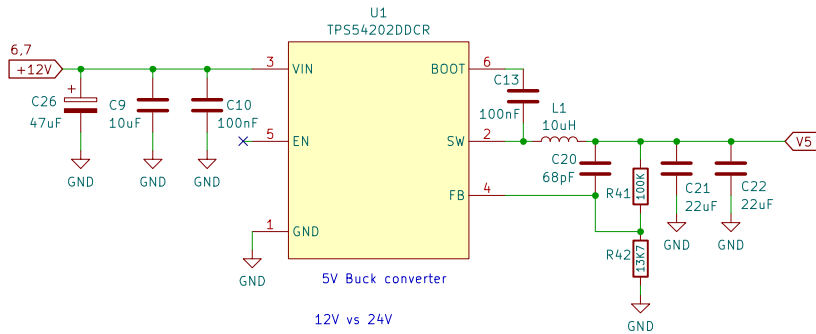
Brookwood Design

Sheet: /Ethernet/
File: Ethernet.kicad_sch

Title: RP23U5XBB

Size: A4 Date: 2025-03-27
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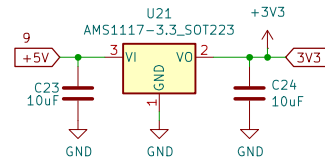
5V Buck converter

12V vs 24V

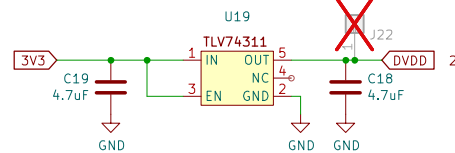
Support 12 to 24V as input. Need to make sure docs are clear on how it works and user precautions.

To do:

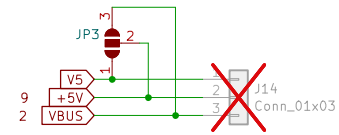
- all LED and opto resistors are properly size with load/power calcs. Check LED illumination at both voltages.
- verify max voltage on all caps.
- verify labeling is clear (call it 12/24 and document 24V compat?)
- Test to 10% over-voltage.



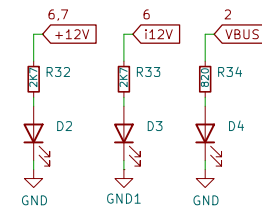
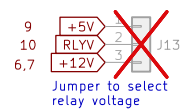
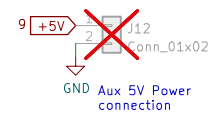
3.3V LDO Regulator



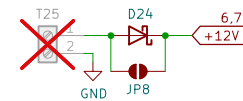
DVDD (1.1V) LDO Regulator



5V bypass to allow USB VBUS +5V for testing and current monitoring. Remove for production PCB? Convert to solder jumper?



Power Indicators



Upper board 12/24V Input

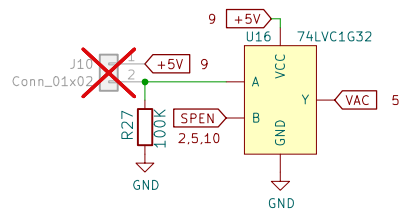
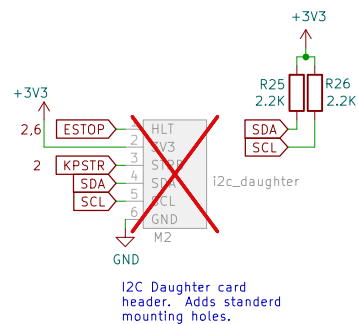
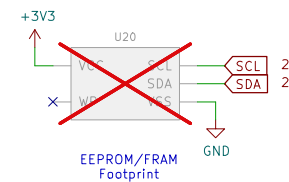
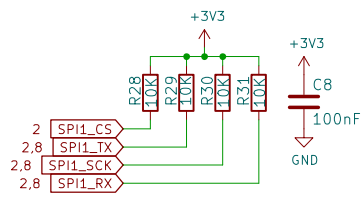
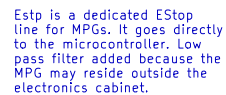
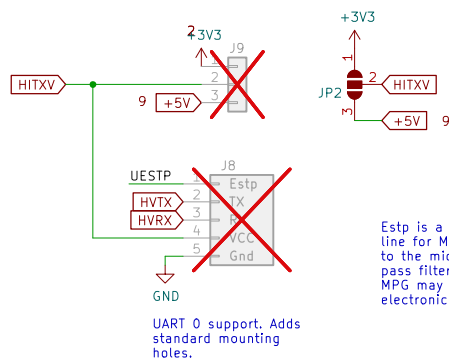
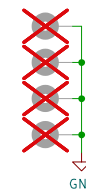
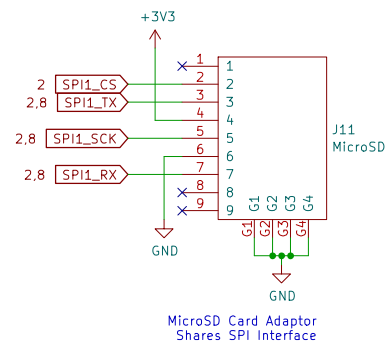
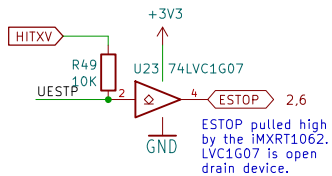
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Sheet: /Power/
File: power.kicad_sch

Title: RP23U5XBB

Size: A4 Date: 2025-03-27
KiCad E.D.A. 8.0.9

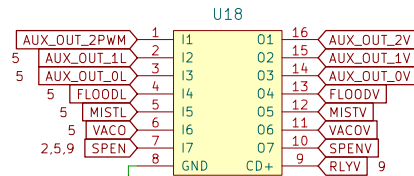
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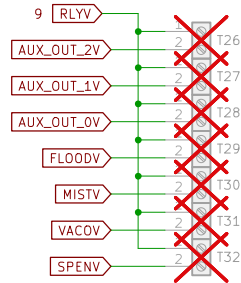
Independent Control for Dust Extractor. A switch attached to the pin header can turn on the DE separately from the spindle.

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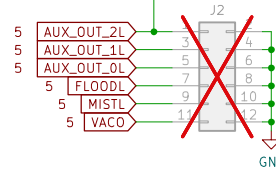
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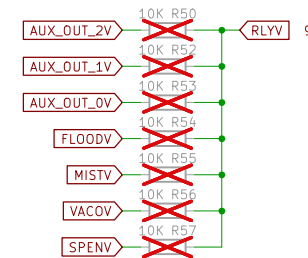
Relay Driver. Currently using ULN2003. Looking at using TBD620003 MOSFET Driver.



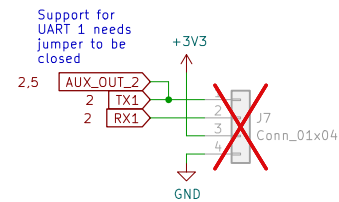
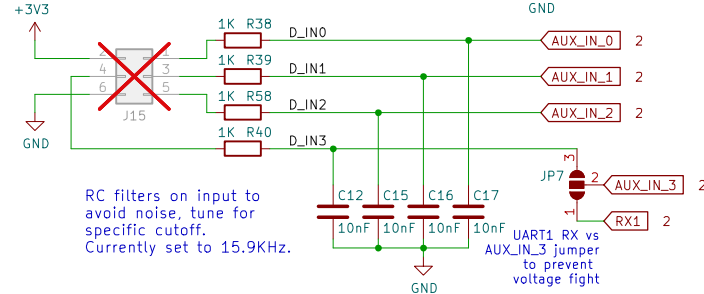
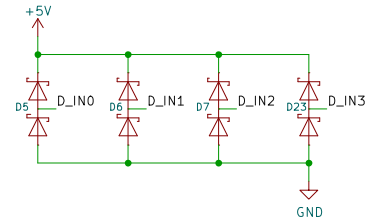
Jumper for Aux_Out_2 to be selected between spindle PWM and digital output. Allows driving 12V PWM loads (like lasers).



All these outputs have HC level drive capability, limited to 6mA.



Digital input group. 3 inputs are 5V tolerant, diode protected with schmitt triggers. RC Low Pass noise filters.



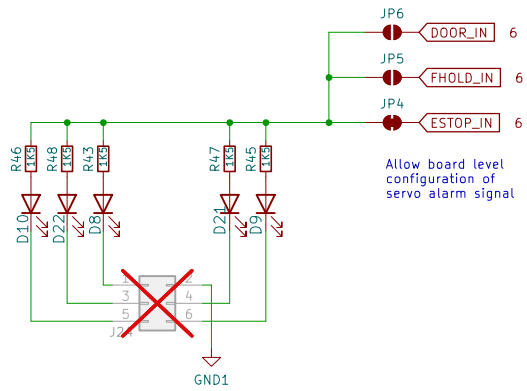
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Sheet: /DigitalIO and Relays/
File: DigitalIO and Relays.kicad_sch

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Sheet: /Servo_error/
File: servo_error.kicad_sch

Title: RP23U5XBB

Size: A4 Date: 2025-03-27
KiCad E.D.A. 8.0.9

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